Ολοκληρώματα

ΤΥΠΟΛΟΓΙΟ

Πολυωνυμικές συναρτήσεις

1. $\int c\,dx = cx + c$
2. $\int x\,dx = \frac{x^2}{2} + c$
3. $\int x^\nu\,dx = \frac{x^{\nu+1}}{\nu+1} + c$
4. $\int (\beta + ax)\,dx = \beta x + \frac{ax^2}{2} + c$
5. $\int (a+x)^2\,dx = a^2x + ax^2 + \frac{x^3}{3} + c$
6. $\int (a+x)^\nu\,dx = \frac{(a+x)^{\nu+1}}{\nu+1} + c$

Ρητές συναρτήσεις

7. $\int \frac{1}{x}\,dx = \log(x) + c$
8. $\int \frac{1}{\beta+ax}\,dx = \frac{\log(\beta+ax)}{a} + c$
9. $\int \frac{1}{(\beta+ax)^2}\,dx = -\frac{1}{a(\beta+ax)} + c$
10. $\int \frac{1}{(\beta+x)(a+x)}\,dx = \frac{-\log(\beta+x)+\log(a+x)}{\beta-a} + c$
11. $\int \frac{x}{\beta+ax}\,dx = \frac{-\beta\log(\beta+ax)+ax}{a^2} + c$
12. $\int \frac{1}{x^2+1}\,dx = \operatorname{atan}(x) + c$
13. $\int \frac{1}{a^2+x^2}\,dx = \frac{\operatorname{atan}\left(\frac{x}{a}\right)}{a} + c$
14. $\int \frac{x}{a^2+x^2}\,dx = \frac{\log\left(a^2+x^2\right)}{2} + c$
15. $\int \frac{x^2}{a^2+x^2}\,dx = -a\operatorname{atan}\left(\frac{x}{a}\right) + x + c$
16. $\int \frac{1}{x^2-1}\,dx = \frac{\log(x-1)}{2} - \frac{\log(x+1)}{2} + c$
17. $\int \frac{1}{-a^2+x^2}\,dx = \frac{\log(-a+x)-\log(a+x)}{2a} + c$
18. $\int \frac{x^3}{x^2-1}\,dx = \frac{x^2}{2} + \frac{\log\left(x^2-1\right)}{2} + c$
19. $\int \sqrt{x}\,dx = \frac{2x^{\frac{3}{2}}}{3} + c$
20. $\int \sqrt{1-x^2}\,dx = \frac{x\sqrt{1-x^2}}{2} + \frac{\operatorname{asin}(x)}{2} + c$
21. $\int \frac{1}{\sqrt{\beta+ax}}\,dx = \frac{2\sqrt{\beta+ax}}{a} + c$

Τριγωνομετρικές συναρτήσεις

22. $\int \sin(x)\,dx = -\cos(x) + c$
23. $\int \cos(x)\,dx = \sin(x) + c$
24. $\int \tan(x)\,dx = -\log(\cos(x)) + c$
25. $\int \cot(x)\,dx = \log(\sin(x)) + c$
26. $\int \frac{1}{\cos^2(x)}\,dx = \tan(x) + c$
27. $\int \frac{1}{\sin^2(x)}\,dx = -\frac{1}{\tan(x)} + c$
28. $\int \sinh(x)\,dx = \cosh(x) + c$
29. $\int \cosh(x)\,dx = \sinh(x) + c$
30. $\int \tanh(x)\,dx = x - \log(\tanh(x)+1) + c$
31. $\int \coth(x)\,dx = x - \log(\tanh(x)+1) + \log(\tanh(x)) + c$
32. $\int \sin(ax)\,dx = -\frac{\cos(ax)}{a} + c$
33. $\int \cos(ax)\,dx = \frac{\sin(ax)}{a} + c$
34. $\int \tan(ax)\,dx = -\frac{\log(\cos(ax))}{a} + c$
35. $\int \cot(ax)\,dx = \frac{\log(\sin(ax))}{a} + c$
36. $\int \sin^2(x)\,dx = \frac{x}{2} - \frac{\sin(2x)}{4} + c$
37. $\int \cos^2(x)\,dx = \frac{x}{2} + \frac{\sin(2x)}{4} + c$
38. $\int \sin(x)\cos(x)\,dx = \frac{\sin^2(x)}{2} + c$
39. $\int \sinh(x)\cosh(x)\,dx = \frac{\cosh^2(x)}{2} + c$
40. $\int \frac{1}{\sin(x)}\,dx = \frac{\log(\cos(x)-1)}{2} - \frac{\log(\cos(x)+1)}{2} + c$
41. $\int \frac{1}{\cos(x)}\,dx = -\frac{\log(\sin(x)-1)}{2} + \frac{\log(\sin(x)+1)}{2} + c$
42. $\int \frac{1}{\sin^2(x)}\,dx = -\frac{1}{\tan(x)} + c$
43. $\int \frac{1}{\cos^2(x)}\,dx = \tan(x) + c$
44. $\int x\sin(x)\,dx = -x\cos(x) + \sin(x) + c$

Εκθετικές - Λογαριθμικές συναρτήσεις

45. $\int e^x\,dx = e^x + c$
46. $\int e^{ax}\,dx = \frac{e^{ax}}{a} + c$
47. $\int \log(x)\,dx = x(\log(x)-1) + c$
48. $\int \log(ax)\,dx = x(\log(ax)-1) + c$
49. $\int e^{\beta+ax}\,dx = \frac{e^{\beta+ax}}{a} + c$
50. $\int xe^x\,dx = (x-1)e^x + c$
51. $\int x^2e^x\,dx = \left(x^2-2x+2\right)e^x + c$
52. $\int x\log(x)\,dx = \frac{x^2(2\log(x)-1)}{4} + c$
53. $\int x^2\log(x)\,dx = \frac{x^3(3\log(x)-1)}{9} + c$
54. $\int \frac{e^x}{x}\,dx = \operatorname{Ei}(x) + c$
55. $\int \frac{\log(x)}{x}\,dx = \frac{\log(x)^2}{2} + c$
56. $\int e^x\sin(x)\,dx = -\frac{\sqrt{2}e^x\cos\left(x+\frac{\pi}{4}\right)}{2} + c$
57. $\int e^x\cos(x)\,dx = \frac{\sqrt{2}e^x\sin\left(x+\frac{\pi}{4}\right)}{2} + c$